Will start at 9:31 am :)

ECON1050 Tutorial 2

Summation Notation & Solving Equations

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Today's Plan

- Recap: core concepts from Lecture 2
- Tutorial questions: Q1–Q5

Recap: Summation notation

A sequence and its sum

For a sequence a_i, summation notation is

$$\sum_{i=m}^{N} a_i.$$

example: a_1, a_2, a_3, a_4

$1, 2, 3, 4$

$a_1 + a_2 + a_3 + a_4 = \sum_{i=1}^{4} a_i$

It represents the sum of the terms a_i as the index i runs from the lower limit m to the upper limit N.

Terminology

- Σ: summation symbol
- a_i: summand
- i: index of summation
- m: lower limit of summation
- N: upper limit of summation

Recap: Rules of summation

Additivity

$$\sum_{i=1}^{N}(a_i + b_i) = \sum_{i=1}^{N} a_i + \sum_{i=1}^{N} b_i.$$

Homogeneity

For a constant c,

$$\sum_{i=1}^{N}(ca_i) = c\sum_{i=1}^{N} a_i.$$

Sum of a constant

$$\sum_{i=1}^{N} c = Nc.$$

Recap: Double sums

Notation

A double sum has two indices, for example

example: $\sum_{i=1}^{2}\sum_{j=1}^{3} a_{ij}$

$$\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}.$$

$i \backslash j$	1	2	3
1	a_{11}	a_{12}	a_{13}
2	a_{21}	a_{22}	a_{23}

How to evaluate

Evaluate the inner sum first for each value of the outer index, and then add the resulting totals.

$$\sum_{i=1}^{m}\left(\sum_{j=1}^{n} a_{ij}\right).$$

Recap: Equations, variables, constants and parameters

example : $X^2 + X = 5$

Equations

An equation describes a relationship between variables, or between variables and constants. An equation can have a unique solution, multiple solutions, or no solution.

Variables and constants

- A **variable** is a quantity whose magnitude may change.
- Variables may be endogenous/dependent or exogenous/independent/predetermined.
- A **constant** has a fixed value.
- A constant multiplying a variable is its **coefficient**.
- A symbolic constant whose value may differ across models, but is fixed within a model, is a **parameter**.

Recap: Equivalent equations

Solving an equation

To solve an equation means to find all values of the variables for which the equation is satisfied.

Operations that preserve equivalence

Equivalent equations have the same solutions. The following operations preserve equivalence:

- add the same number to both sides;
- multiply both sides by the same non-zero number;
- replace either side by an equal expression.

example: $\frac{1}{2}x^2 - 1 = 7$ ① proposition

$\Leftrightarrow \frac{1}{2}x^2 = 8$ ② proposition

$\Leftrightarrow x^2 = 16$ ③

$\Rightarrow x = \pm 4$ ④

Notation

Equivalent statements may be connected by

$$\Leftrightarrow .$$

Recap: Implications when solving equations

Operations that may not preserve equivalence

The following operations can change the solution set:

- multiplying or dividing both sides by an expression involving a variable;
- raising both sides to equal powers, including taking roots.

Implication

In these cases, use the implication arrow

$$\Rightarrow .$$

An equivalence can be replaced by an implication, but an implication cannot automatically be replaced by an equivalence. Candidate solutions may need to be checked in the original equation.

Recap: Quadratic equations

General form

$$ax^2 + bx + c = 0, \qquad a \neq 0.$$

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

For real-valued solutions, the discriminant must satisfy

$$b^2 - 4ac \geq 0.$$

Other methods

Quadratic equations can also be solved by factoring or by completing the square.

Recap: Roots and factorisation

Factorisation from roots

If x_1 and x_2 are the roots of

$$ax^2 + bx + c = 0,$$

then

$$ax^2 + bx + c = a(x - x_1)(x - x_2).$$

Root relationships

For

$$x^2 + px + q = 0,$$

the roots satisfy

$$x_1 + x_2 = -p, \qquad x_1 x_2 = q.$$

Recap: Nonlinear equations

General idea

For more complex nonlinear equations, full or partial solutions can sometimes be obtained using the same algebraic techniques used for simpler equations.

Important caution

If a factor is cancelled, check that the cancelled factor is not equal to zero. Otherwise, valid solutions may be lost or invalid steps may be introduced.

Example: $x^3 + 2x^2 + x = 0$

$x(x^2 + 2x + 1) = 0$

$x(x+1)^2 = 0$

$x = 0, -1$

Recap: Systems of two linear equations

Two equations in two unknowns

A system of two linear equations can be written as

$$\begin{cases} ax + by = c, & ① \\ dx + ey = f. & ② \end{cases}$$

①: $ax = c - by$

$x = \frac{c-by}{a}$ ③

plug ③ into ②: $d \cdot \frac{c-by}{a} + ey = f$

$y = \frac{af-cd}{ae-bd}$ ④

Two standard solution methods are **substitution** and **elimination**. plug ④ into ③: $x = \frac{ce-bf}{ae-bd}$

Unique solution

If $ae \neq bd$, then

$$x = \frac{ce - bf}{ae - bd}, \qquad y = \frac{af - cd}{ae - bd}.$$

Tutorial Q1

Question 1

Evaluate the following sums.

Finite terms

$\Rightarrow$ write down all the term

$\Rightarrow$ add one by one

(a)

$$\sum_{i=1}^{4} \frac{1}{i(i+2)}.$$

(b)

$$\sum_{j=5}^{9} (2j-8)^2.$$

(c)

$$\sum_{s=0}^{2} \sum_{r=2}^{4} \left(\frac{rs}{r+s} \right)^2.$$

(a) $\sum_{i=1}^{4} \frac{1}{i(i+2)} = \underset{i=1}{\frac{1}{1(1+2)}} + \underset{i=2}{\frac{1}{2(2+2)}} + \underset{i=3}{\frac{1}{3(3+2)}} + \underset{i=4}{\frac{1}{4(4+2)}}$

$= \frac{1}{3} + \frac{1}{8} + \frac{1}{15} + \frac{1}{24}$ Calculator

$= \frac{17}{30}$

(b) $\sum_{j=5}^{9} (2j-8)^2 = (2\times5-8)^2 + (2\times6-8)^2 + (2\times7-8)^2 + (2\times8-8)^2 + (2\times9-8)^2$

$= 4 + 16 + 36 + 64 + 100$

$= 220$

(c) $\sum_{s=0}^{2} \sum_{r=2}^{4} \left(\frac{rs}{r+s}\right)^2 = \left(\frac{2\times0}{2+0}\right)^2 + \left(\frac{2\times1}{2+1}\right)^2 + \left(\frac{2\times2}{2+2}\right)^2 + \left(\frac{3\times0}{7+0}\right)^2 + \left(\frac{3\times1}{3+1}\right)^2 + \left(\frac{3\times2}{3+2}\right)^2 + \left(\frac{4\times0}{4+0}\right)^2 + \left(\frac{4\times1}{4+1}\right)^2 + \left(\frac{4\times2}{4+2}\right)^2$

$= \frac{21113}{3600}$

r \ s	0	1	2
2	$\left(\frac{2\times0}{2+0}\right)^2$	$\left(\frac{2\times1}{2+1}\right)^2$	$\left(\frac{2\times2}{2+2}\right)^2$
3	$\left(\frac{3\times0}{7+0}\right)^2$	$\left(\frac{3\times1}{3+1}\right)^2$	$\left(\frac{3\times2}{3+2}\right)^2$
4	$\left(\frac{4\times0}{4+0}\right)^2$	$\left(\frac{4\times1}{4+1}\right)^2$	$\left(\frac{4\times2}{4+2}\right)^2$

Tutorial Q1: Knowledge used

Knowledge used

- Interpret the lower limit, upper limit, index and summand in Σ notation.
- Evaluate a finite sum by substituting each index value into the summand.
- For a double sum, evaluate the inner sum first and then the outer sum.
- The additivity rule allows a sum of several terms to be separated into several sums.

Tutorial Q1(a): Evaluate the sum

Solution

$$\sum_{i=1}^{4} \frac{1}{i(i+2)} = \frac{1}{1(1+2)} + \frac{1}{2(2+2)} + \frac{1}{3(3+2)} + \frac{1}{4(4+2)}.$$

Hence

$$= \frac{1}{3} + \frac{1}{8} + \frac{1}{15} + \frac{1}{24} = \frac{204}{360} = \frac{17}{30}.$$

Tutorial Q1(b): Evaluate the sum

Solution

$$\begin{aligned}\sum_{j=5}^{9}(2j-8)^2 &= (2\cdot 5-8)^2+(2\cdot 6-8)^2+(2\cdot 7-8)^2\\&\quad+(2\cdot 8-8)^2+(2\cdot 9-8)^2\\&=4+16+36+64+100\\&=220.\end{aligned}$$

Tutorial Q1(c): Evaluate the double sum

Solution

$$
\begin{aligned}
&\sum_{s=0}^{2}\sum_{r=2}^{4}\left(\frac{rs}{r+s}\right)^2 \\
&= \sum_{s=0}^{2}\left[\left(\frac{2s}{2+s}\right)^2 + \left(\frac{3s}{3+s}\right)^2 + \left(\frac{4s}{4+s}\right)^2\right] \\
&= \sum_{s=0}^{2}\left(\frac{2s}{2+s}\right)^2 + \sum_{s=0}^{2}\left(\frac{3s}{3+s}\right)^2 + \sum_{s=0}^{2}\left(\frac{4s}{4+s}\right)^2 \\
&= 0 + \left(\frac{2}{3}\right)^2 + \left(\frac{4}{4}\right)^2 + 0 + \left(\frac{3}{4}\right)^2 + \left(\frac{6}{5}\right)^2 + 0 + \left(\frac{4}{5}\right)^2 + \left(\frac{8}{6}\right)^2 \\
&= 5 + \frac{3113}{3600}.
\end{aligned}
$$

Tutorial Q2

Question 2

Express the following sums in sum notation.

5 mins

(a)

$$3 + 5 + 7 + \cdots + 199 + 201.$$

$\sum_{i=m}^{N} a_i$

① $a_i = ?$

② $m = ?$

③ $N = ?$

(b)

$$\frac{2}{1} + \frac{3}{2} + \frac{4}{3} + \cdots + \frac{97}{96}.$$

(c)

$$1 + \frac{x^2}{3} + \frac{x^4}{5} + \frac{x^6}{7} + \cdots + \frac{x^{32}}{33}.$$

(a) $3+5+7+\dots+199+201$

① odd numbers: $a_i = \boxed{2i+1}$ or $2i-1$

② m: $3 = 2m+1 \Rightarrow m=1$

③ n: $201 = 2n+1 \Rightarrow n=100$

$$\Rightarrow \sum_{i=1}^{100}(2i+1)$$

(b) $\frac{2}{1}+\frac{3}{2}+\frac{4}{3}+\dots+\frac{97}{96}$

① $a_i = \frac{i+1}{i}$

② m: $\frac{2}{1} = \frac{m+1}{m} \Rightarrow m=1$

③ n: $\frac{97}{96} = \frac{n+1}{n} \Rightarrow n=96$

$$\Rightarrow \sum_{i=1}^{96}\frac{i+1}{i}$$

(c) $1+\frac{x^2}{3}+\frac{x^4}{5}+\frac{x^6}{7}+\dots+\frac{x^{32}}{33}$

even number : $2i$

odd number : $2i+1$

① $a_i = \frac{x^{2i}}{2i+1}$

② m: $1 = \frac{x^{2m}}{2m+1} \Rightarrow m=0$

③ n: $\frac{x^{32}}{33} = \frac{x^{2n}}{2n+1} \Rightarrow n=16$

$$\Rightarrow \sum_{i=0}^{16}\frac{x^{2i}}{2i+1}$$

Tutorial Q2: Knowledge used

Knowledge used

- Identify a formula for the general term of a sequence.
- Choose an index and determine the corresponding lower and upper limits.
- Different index choices can give equivalent summation expressions.
- Match numerator and denominator patterns separately when forming the summand.

Tutorial Q2(a): Sum notation

Solution

The terms are odd numbers, so use $2i - 1$.

$$3 = 2(2) - 1, \qquad 201 = 2(101) - 1.$$

Therefore,

$$\sum_{i=2}^{101} (2i - 1).$$

An equivalent indexing is

$$\sum_{i=1}^{100} (2i + 1).$$

Tutorial Q2(b): Sum notation

Solution

The general term is

$$\frac{i+1}{i}.$$

Indeed,

$$\frac{2}{1} = \frac{i+1}{i} \text{ for } i = 1, \qquad \frac{97}{96} = \frac{i+1}{i} \text{ for } i = 96.$$

Thus,

$$\sum_{i=1}^{96} \frac{i+1}{i}.$$

Tutorial Q2(c): Sum notation

Solution

The exponent in the numerator follows the even-number sequence $2i$, while the denominator follows the odd-number sequence $2i+1$.

$$\frac{x^{2i}}{2i+1} = 1 \quad \text{when } i = 0, \qquad \frac{x^{2i}}{2i+1} = \frac{x^{32}}{33} \quad \text{when } i = 16.$$

Therefore,

$$\sum_{i=0}^{16} \frac{x^{2i}}{2i+1}.$$

Tutorial Q3

Question 3

Solve the following equations.

5 mins

(a)

$$\frac{6x}{5} - \frac{5}{x} = \frac{2x-3}{3} + \frac{8x}{15}.$$

(b)

$$\frac{\sqrt{x+10}}{\sqrt{x+3}} = \sqrt{x-2}.$$

(a) $\frac{6x}{5} - \frac{5}{x} = \frac{2x-3}{3} + \frac{8x}{15}$ fraction $\Rightarrow$ integer

$$\frac{6x}{5}\cdot 15x - \frac{5}{x}15x = \frac{(2x-3)}{3}\cdot 15x + \frac{8x}{15}\cdot 15x$$

$$18x^2 - 75 = (2x-3)\cdot 5x + 8x^2$$

$$18x^2 - 75 = 10x^2 - 15x + 8x^2$$

$$-75 = -15x$$

$$x = \frac{-75}{-15} = 5$$

(b) $\frac{\sqrt{x+10}}{\sqrt{x+3}} = \sqrt{x-2}$ Square root $\begin{cases} x+10 \geq 0 \Leftrightarrow x \geq -10 \\ x+3 \geq 0 \Leftrightarrow x \geq -3 \\ x-2 \geq 0 \Leftrightarrow x \geq 2 \end{cases}$ $\Rightarrow x \geq 2$

$$\left(\frac{\sqrt{x+10}}{\sqrt{x+3}}\right)^2 = (\sqrt{x-2})^2$$

$$\frac{x+10}{x+3} = x-2$$

$$x+10 = (x-2)(x+3)$$

$$x+10 = x^2-2x+3x-6$$

$$x^2 = 16$$

$$x = \pm 4$$

$$x = 4 \quad (x \geq 2)$$

Tutorial Q3: Knowledge used

Knowledge used

- Use algebraic operations to obtain simpler equivalent equations when the operations preserve the solution set.
- Expressions in denominators must be non-zero.
- Multiplying by an expression involving a variable requires attention to restrictions on that expression.
- Squaring both sides may produce only an implication rather than an equivalence, so candidate solutions must be checked in the original equation.

Tutorial Q3(a): Solve the rational equation

Solution

$$\frac{6x}{5} - \frac{5}{x} = \frac{2x-3}{3} + \frac{8x}{15}$$

$$\frac{6x^2 - 25}{5x} = \frac{5(2x-3) + 8x}{15}.$$

Hence

$$15(6x^2 - 25) = 5x(18x - 15),$$

$$90x^2 - 375 = 90x^2 - 75x,$$

$$75x = 375.$$

Therefore,

$$x = 5.$$

Tutorial Q3(b): Square both sides

Solution

Starting from

$$\frac{\sqrt{x+10}}{\sqrt{x+3}} = \sqrt{x-2},$$

square both sides:

$$\frac{x+10}{x+3} = x-2.$$

Then

$$x+10 = (x-2)(x+3) = x^2 + x - 6,$$

so

$$x^2 = 16 \quad \Rightarrow \quad x = -4 \text{ or } x = 4.$$

Because we squared both sides, re-check the candidates in the original equation. $x = -4$ does not work, while $x = 4$ gives both sides equal to $\sqrt{2}$.
Therefore,

$$x = 4.$$

Question 4

Solve the following quadratic equations.

(a) Factor

$$12p^2 - 7p + 1 = 0.$$

(b) Solve

$$42 = x^2 + x$$

by completing the square.

① Factorization direct

② two solutions
$\Rightarrow a(p-p_1)(p-p_2)$

$ax^2+bx+c=0 \Rightarrow x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

(a) $12p^2 - 7p + 1 = 0$. Here $a=12$, $b=-7$, $c=1$

$\Rightarrow p = \frac{7 \pm \sqrt{49 - 4\times 12 \times 1}}{2\times 12} = \frac{7\pm 1}{24} \Rightarrow \begin{cases} p_1 = \frac{8}{24} = \frac{1}{3} \\ p_2 = \frac{6}{24} = \frac{1}{4} \end{cases}$

$\Rightarrow 12p^2 - 7p + 1 = 12(p - \frac{1}{4})(p - \frac{1}{3})$

(b) $x^2 + x = 42$ $\quad 42 = \frac{168}{4}$

$x^2 + x + \frac{1}{4} = 42 + \frac{1}{4}$

$(x + \frac{1}{2})^2 = \frac{169}{4}$

$x + \frac{1}{2} = \pm \frac{13}{2}$

$x = \pm \frac{13}{2} - \frac{1}{2}$

$\Rightarrow x_1 = \frac{13-1}{2} = 6$

$x_2 = \frac{-13-1}{2} = -7$

$(x+a)^2 = x^2 + 2ax + a^2$

$1 = 2a \Rightarrow a = \frac{1}{2}$

$\Rightarrow a^2 = \frac{1}{4}$

Tutorial Q4: Knowledge used

Knowledge used

- A quadratic equation has the form $ax^2 + bx + c = 0$ with $a \neq 0$.
- Use the quadratic formula to find roots when needed.
- If the roots are known, factor using $a(x - x_1)(x - x_2)$.
- Completing the square uses the identity $x^2 + 2ax + a^2 = (x + a)^2$.

Tutorial Q4(a): Roots and factorisation

Solution

For

$$12p^2 - 7p + 1 = 0,$$

the discriminant is

$$(-7)^2 - 4(12)(1) = 1 \geq 0.$$

Thus

$$p_{1,2} = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(12)(1)}}{2(12)} = \frac{7 \pm 1}{24}.$$

Hence

$$p_1 = \frac{1}{4}, \qquad p_2 = \frac{1}{3}.$$

Therefore,

$$12p^2 - 7p + 1 = 12\left(p - \frac{1}{4}\right)\left(p - \frac{1}{3}\right).$$

Tutorial Q4(b): Completing the square

Solution

$$x^2 + x = 42.$$

Since

$$x^2 + 2ax + a^2 = (x + a)^2$$

and the coefficient of x is 1, take $a = \frac{1}{2}$. Add $\frac{1}{4}$ to both sides:

$$x^2 + x + \frac{1}{4} = 42 + \frac{1}{4},$$

$$\left(x + \frac{1}{2}\right)^2 = \frac{169}{4}.$$

Hence

$$x + \frac{1}{2} = \pm\frac{13}{2},$$

so

$$x = -7 \quad \text{or} \quad x = 6.$$

Tutorial Q5

Question 5

Solve the following system of equations:

$$\begin{cases} 3x + 6y = 21, & ① \\ 12x - 5y = -3. & ② \end{cases}$$

① apply general solution

② subs and elimination

①: $3x = 21 - 6y$

$x = 7 - 2y$ ③

plug ③ into ②: $12(7-2y) - 5y = -3$

$84 - 24y - 5y = -3$

$87 = 29y$

$y = \frac{87}{29} = 3$

plug $y=3$ into ③ : $x = 7 - 2 \cdot 3$

$= 7 - 6$

$= 1$

$(x, y) = (1, 3)$

Tutorial Q5: Knowledge used

Knowledge used

- A system of two linear equations in two unknowns can be solved by substitution or elimination.
- **Substitution**: solve one equation for one variable and substitute it into the other equation.
- **Elimination**: multiply equations when needed so that one variable cancels when the equations are added or subtracted.

Tutorial Q5: Substitution

Solution

From

$$3x + 6y = 21,$$

we obtain

$$3x = 21 - 6y \quad \Rightarrow \quad x = 7 - 2y.$$

Substitute into $12x - 5y = -3$:

$$12(7 - 2y) - 5y = -3,$$

$$84 - 29y = -3 \quad \Rightarrow \quad 29y = 87 \quad \Rightarrow \quad y = 3.$$

Then

$$x = 7 - 2(3) = 1.$$

Therefore,

$$x = 1, \qquad y = 3.$$

Tutorial Q5: Elimination

Solution

Multiply the first equation by -4:

$$-12x - 24y = -84.$$

Add this to

$$12x - 5y = -3:$$

$$-29y = -87 \quad \Rightarrow \quad y = 3.$$

Substitute into the first equation:

$$3x + 6(3) = 21 \quad \Rightarrow \quad 3x = 3 \quad \Rightarrow \quad x = 1.$$

Thus,

$$x = 1, \qquad y = 3.$$

Key takeaways

- Summation notation provides a compact way to write and evaluate finite sums, including double sums.
- When solving equations, distinguish equivalence from implication and check candidates after non-reversible operations.
- Quadratic equations can be solved by the quadratic formula, factorisation, or completing the square.
- Systems of two linear equations can be solved by substitution or elimination.